

Algorithmics	Student information	Date	Number of session
	UO: 300737	03/02/2025	0
	Surname: Hoyos	 Escuela de Ingeniería Informática Universidad de Oviedo	
	Name: Ignacio		



Activity 1. Factor 1: Problem Size

n	Time (ms)
10000	1603
20000	6483
40000	25671
80000	OoT
160000	OoT
320000	OoT
640000	OoT

Activity 2: Computer performance

PC1 -> CPU: 12th Gen Intel(R) Core(TM) i5-12400 2.50 GHz, RAM: 16 GB

PC2 -> CPU: Ryzen 5 2500U with Radeon Vega Mobile Gfx 2.00GHz, RAM: 8 GB

n	PC1 time (ms)	PC2 time (ms)
10000	1603	3940
20000	6483	15766
40000	25671	OoT
80000	OoT	OoT
160000	OoT	OoT
320000	OoT	OoT
640000	OoT	OoT

Activity 3: Implementation environment

n	Java time (ms)	Python time (ms)
10000	1242	1603
20000	4997	6483
40000	20588	25671
80000	OoT	OoT
160000	OoT	OoT
320000	OoT	OoT
640000	OoT	OoT

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Activity 4 : Algorithm that is used

n	P1	P2	P3
10000	1603	194	0
20000	6483	707	0
40000	25671	2675	0
80000	OoT	10174	0
160000	OoT	37536	0
320000	OoT	OoT	0
640000	OoT	OoT	0

WITHOUT OPTIMIZATION				WITH OPTIMIZATION			
n	JavaA1	JavaA2	JavaA3		JavaA1	JavaA2	JavaA3
10000	2495	79	0		1229	39	0
20000	10171	285	0		5152	133	0
40000	39601	1043	0		19898	485	0
80000	OoT	3767	0		OoT	1850	0
160000	OoT	14513	0		OoT	6932	0
320000	OoT	56305	0		OoT	25918	0
640000	OoT	OoT	1		OoT	OoT	1

CONCLUSION

It is clearly visible that Java optimized programs outperform the non-optimized. As expected, the java Virtual machine refine the code and improve its efficiency. Nevertheless I am quite surprise that in some cases the Java programs without optimization are faster than the python ones, I thought that having the virtual machine but losing its code improvements will be like an extra layer to our code to be process delaying it and increasing the total time. However thin only happens sometimes, so I were partially wrong, I suppose that maybe the "Xint" command totally deactivates the Java Virtual Machine and that way no extra time in added, but I don't know why in some cases like 2 it's faster.